

Corrective Action Report — CAR-2025-005

Field	Value
Document ID	CAR-2025-005
Issue Date	2025-09-09
Author	R. Mendelsohn, Substation Reliability Engineer; reviewed by D. Pham, Compliance Lead
CAR Status	Pending Engineering Review
Subject	ABB Substation Service Manual SE-7100 §6.4 Recommends 4-Year Maintenance Cycle; Cascadia Standard MS-2025-03 §8.1 Requires 2-Year; Conflict Implicates Repeat Failures at SUB-10641
Affected Region / Site / Asset	NW-PowerPool / NW-PowerPool-WIND-014 / SUB-10641 (primary), SUB-10629, SUB-10650
Severity	High
Owner	E. Linhart, Principal Engineer — Rotating Equipment (acting Substation lead); secondary owner P. Iturbe, Substation Reliability Manager
Original Target Close Date	2025-12-31 (missed)
Revised Target Close Date	2026-04-30
Distribution	Engineering Standards Committee; NW-PowerPool Plant Manager; Asset Management; Warranty Administration

1. Finding

ABB Power Grids Substation Service Manual **SE-7100** (filed **VTM-ABB-SE7100**), §6.4 "Recommended Major Maintenance Intervals," specifies a **48-month** (4-year) cycle for medium-voltage breaker overhaul, dielectric oil change, and protective-relay calibration on the 145 kV and 230 kV substation product families deployed at NW-PowerPool sites. Cascadia internal **Standard MS-2025-03 §8.1**, revised February 2025, requires a **24-month** (2-year) cycle for substation assets located in the NW-

PowerPool ambient envelope (winter low <-12 °C, summer high >38 °C, salt-fog exposure within 80 km of the coast).

The 4-year ABB cadence is calibrated against an indoor or temperate outdoor environmental envelope (ABB-published reference: ANSI/IEEE C37.06 normal duty). The NW-PowerPool envelope exceeds the manual's stated envelope on three of four environmental axes. MS-2025-03 §8.1 codifies the more aggressive cycle accordingly.

Maintenance execution at SUB-10641 has historically followed the ABB 4-year cadence. The asset's 15 forced outages in FY2025 — the most of any substation in the Cascadia fleet — appear to be partially attributable to under-maintenance against the environmental-adjusted standard.

2. Evidence

2.1 SUB-10641 maintenance history

Date	Activity	Vendor	Reference
2017-08	Commissioning + initial PM	ABB	Build packet
2021-09	Major overhaul (4-year per SE-7100 §6.4)	ABB Field Service	WO-2021-04412
2025-09	Scheduled 4-year overhaul	—	WO-2025-00549 (deferred per CAR-2025-001)

No intermediate 2-year service per MS-2025-03 §8.1 was scheduled. The 2025 4-year overhaul was deferred under the Q1 budget freeze (cross-reference CAR-2025-001 §2.1, WO-2025-00488, WO-2025-00489, WO-2025-00549).

2.2 SUB-10641 FY2025 outage record

15 forced outages, 13,362 MWh lost, \$801,719 revenue impact. Top events:

- OUT-2025-1632 (2025-11-18, soc_drift, 204 h, 2,252.8 MWh lost) — breaker timing out of tolerance.
- OUT-2025-1627 (2025-06-28, efficiency_decline, 176 h, 1,943.6 MWh lost) — relay miscoordination.
- OUT-2025-1554 (2025-10-03, vibration_excess on adjacent SUB-10605, 81 h, 1,975.1 MWh) — cascading event.
- 12 additional smaller events of which 9 cite breaker-mechanism or relay-calibration drift.

2.3 Comparable substation performance

- SUB-10629 (Hitachi Energy, NW-PowerPool-WIND-024): 4 forced outages, last service 2023-10 (2-year cycle followed). 9,800 MWh lost (heavily weighted by single OUT-2025-1600 event).
- SUB-10650 (ABB, ERCOT-WIND-039): 6 forced outages, on 4-year ABB cycle but with significantly less aggressive ambient envelope.

The SUB-10641 outage rate is approximately 3.75× the SUB-10629 rate within the same region — both ABB and Hitachi Energy assets, the principal procedural variable being the 4-year vs. 2-year cadence.

2.4 ABB warranty interaction

ABB has stated in correspondence dated 2025-07-31 that "service performed by non-ABB-certified technicians on an interval shorter than that recommended in §6.4 does not affect warranty, provided ABB-spec consumables and tools are used." This removes the warranty obstacle to MS-2025-03 compliance. The conflict is therefore not legally constrained; it is operationally and procedurally constrained.

3. Root Cause Hypothesis

- **Primary:** Maintenance plans for SUB-10641 were built in 2017 against SE-7100 §6.4 and have not been retro-fitted to the MS-2025-03 §8.1 standard issued in February 2025.
- **Contributing:** The maintenance-planning team at NW-PowerPool defaults to OEM-manual cadence unless an explicit engineering memo directs otherwise. No such memo was issued to apply MS-2025-03 §8.1 to the existing substation fleet.
- **Contributing:** The 2025 scheduled overhaul was deferred under the budget freeze (CAR-2025-001), removing the natural opportunity to re-baseline the cadence.

4. Conflict With Existing Guidance

- **VTM-ABB-SE7100 §6.4** — 4-year cycle.
- **Cascadia Standard MS-2025-03 §8.1** — 2-year cycle for NW-PowerPool ambient envelope.
- **IEEE C37.10.1-2018 §5.2** (industry consensus standard, breaker condition-based maintenance) — supports environmental-adjusted cadence; references temperature cycling factors that align with MS-2025-03.

This conflict, unlike CAR-2025-002 (Siemens Gamesa), does not have warranty implications. It is purely a cost-and-cadence conflict where the OEM manual recommends less maintenance than internal standards require.

5. Recommended Corrective Action

1. Issue **Substation Maintenance Plan Update SMP-2025-04** retro-fitting MS-2025-03 §8.1 cadence to all NW-PowerPool ABB and Hitachi Energy substation assets effective 2026-Q1.
2. Schedule immediate (within 90 days of CAR re-issue) interim service at SUB-10641 covering breaker timing, oil DGA, and relay calibration. Estimated scope: \$148K (CPS quote, dated 2025-08-30).
3. Add SUB-10641 to the Engineering Review list (cross-reference CAR-2025-009 — escalation policy non-compliance).
4. Update Maintenance Engineering training to clarify that OEM-manual cadence is a **floor**, not a ceiling, when MS-2025-03 specifies more frequent intervals.
5. Add an Engineering co-sign requirement to any maintenance plan that uses OEM-manual default cadence on Tier-1 substation assets.

6. Action Owner & Timeline

Sub-action	Owner	Target Date
SMP-2025-04 published	P. Iturbe	2026-02-28
SUB-10641 interim service	NW-PowerPool maintenance + CPS	2026-01-31
Add SUB-10641 to Engineering Review	E. Linhart	2025-10-15 — partial; on advisory list only
Maintenance Engineering training update	Maintenance Engineering Standards	2026-03-31
CMMS co-sign requirement	R. Vasquez	2026-04-30

7. Implementation Status

Pending Engineering Review. SMP-2025-04 is in draft. The SUB-10641 interim service has been quoted but not approved; procurement is holding pending resolution of CAR-2025-004 (Apex contract issue) since the asset's emergency response would otherwise default to Apex. Another forced outage (OUT-2025-1632) occurred two months after this CAR was issued. The deferred 2025 overhaul (WO-2025-00549) was eventually executed 2025-12-04 at a cost of \$284K, against an original budget estimate of \$112K.

Cross-references: CAR-2025-001 (deferred WO-2025-00488, -00489, -00549 affecting SUB-10641); CAR-2025-009 (SUB-10641 should have been on Engineering Review list after 3 forced outages, was not until outage #11); CAR-2025-002 (parallel OEM-vs-MS-2025-03 conflict pattern); VTM-ABB-SE7100 (OEM source of conflict); Cascadia Standard MS-2025-03 §8.1; OIR-2025-008 (SUB-10641 cumulative outage investigation); EMR-2025-002 (Apex invoice for OUT-2025-1627); RCA-2025-005 (SUB-10641 breaker-timing root cause).

End of report.